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Your grade: 100%

Your latest: 100% • Your highest: 100%

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Learning Check - Basic Probability, Continued

Instructions

Review Learning Objectives

1. Let X be a Random Variable that represents the result of rolling an unweighted 6-sided die. What is $E[X]$? 1 / 1 point

3.5

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practiceLet's chat

Correct

$E[X]$ = sum of $x_i \cdot p(x_i)$ for $i = 1, 6 = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = (1 + 2 + 3 + 4 + 5 + 6) / 6 = 21 / 6 = 3.5$

Due

Sep 30, 11:59 PM IST

Attempts

3 left (5 attempts every 24 hours)

Try again

Submitted

Sep 27, 1:13 AM IST

2. Let X be a Random Variable that represents the result of rolling an unweighted 6-sided die. What is $\text{Var}(X)$? 1 / 1 point

35/12

75/12

7/2

35/2

Your grade

To pass you need at least 80%. We keep your highest score.

View submission

See feedback

100%

Correct

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$\text{Var}(X) = E[(x - E[X])^2]$ = sum of $p(x_i) \cdot (x_i - E[X])^2 = \frac{1}{6}(1 - 3.5)^2 + \frac{1}{6}(2 - 3.5)^2 + \frac{1}{6}(3 - 3.5)^2 + \frac{1}{6}(4 - 3.5)^2 + \frac{1}{6}(5 - 3.5)^2 + \frac{1}{6}(6 - 3.5)^2 = \frac{1}{6}(\frac{5}{2})^2 + \frac{1}{6}(\frac{5}{2})^2 + \frac{1}{6}(\frac{5}{2})^2 + \frac{1}{6}(\frac{5}{2})^2 + \frac{1}{6}(\frac{5}{2})^2 + \frac{1}{6}(\frac{5}{2})^2 = \frac{1}{6} \cdot \frac{70}{4} = \frac{70}{24} = \frac{35}{12}$

3. Let X be a random variable, and X_1, X_2 be i.i.d. copies of X . 1 / 1 point

Which of the following does not have a lower variance than X ?

$\frac{1}{2}X$

$\frac{X_1 + X_2}{2}$

X^2

Correct

https://www.coursera.org/learn/statistics-for-data-science-essentials/assignment-submission/1mrfy/learning-check-basic-probability-continued/view-feedback

1/1